Lightweight Hyperspectral Domain Generalization via Sample-Specific Low-Rank Basis Mixture

Jialiang Qiu, Mengqi Chai, Shuming Xiong, and Qiang Zhou

***Abstract*—Hyperspectral image (HSI) classification often suffers from substantial cross-domain shifts arising from sensor heterogeneity, varying acquisition conditions, and scene-dependent spectral distortions—all of which significantly degrade model robustness in previously unseen environments. Moreover, most existing deep learning approaches rely on heavy architectures, hindering their deployment on resource-constrained platforms such as UAVs or onboard processors. To address these challenges, we propose the adaptive basis spectral network (ABSNet), a lightweight domain generalization (DG) framework that learns adaptive low-rank spectral bases for constructing sample-specific low-rank adapters. By integrating spectral–spatial representations through cost-efficient token modulation, ABSNet is designed to improve robustness to distribution shifts while retaining discriminative spectral information. In addition, the rank-constrained formulation retains representation capacity with minimal parameter overhead, making the framework suitable for edge-level deployment. Experiments on the Houston, Pavia, and HyRANK benchmarks show that ABSNet achieves accuracy comparable to state-of-the-art (SOTA) methods with only a small fraction of the parameters of the compared methods. Ablation studies assess the contribution of each component within the proposed framework. The code will be made publicly available at https://github.com/qiujialiang/ABSNet**

***Index Terms*—Domain generalization (DG), hyperspectral image (HSI) classification, lightweight neural network, low-rank adaptation, spectral–spatial feature fusion.**

I. Introduction

HYPERSPECTRAL image (HSI) data have become indispensable in Earth observation due to their ability to capture rich spectral signatures across hundreds of contiguous bands. This capability enables fine-grained material discrimination in applications such as land-cover mapping [1], precision agriculture [2], environmental monitoring [3], and disaster response [4]. However, obtaining sufficient pixel-level labels is costly and labor-intensive, so only limited labeled samples are often available [5], [6]. As a result, only a few source scenes are fully annotated, while most target scenes remain unlabeled. Supervised models trained on such limited source data often fail when deployed in unseen environments exhibiting domain shifts caused by sensor discrepancies, atmospheric conditions, illumination variations, and geographical diversity [7], [8].

Received 6 April 2026; revised 3 August 2026; accepted 21 September 2026. Date of publication 2 October 2026; date of current version 19 October 2026. This work was supported by Youth Fund Project of Jiangsu Provincial Basic Research Program under Grant BK20240837. *(Corresponding author: Qiang Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2026.3739820

Domain generalization (DG) addresses this challenge by learning representations that remain robust under cross-domain variations, enabling generalization to unseen target scenes without target supervision [9], [10]. Although DG has shown promise in vision and remote sensing, applying it to HSI data remains challenging because differences in acquisition conditions, scenes, and sensors induce substantial cross-domain distribution shifts [8], [11]. Variations in sensor characteristics and environmental conditions can induce substantial spectral and distribution discrepancies across scenes [8].

Recent work on cross-domain HSI classification includes both domain-adaptation (DA) and DG methods, which employ diverse strategies to address domain shifts. In the DA setting, adaptive smooth adversarial learning uses adversarial alignment and smoothing regularization to improve cross-domain classification [12]. On the contrary, a GAN-based domain-generation framework learns domain-invariant representations from source data for unseen-DG [13]. A shift-reduction network directly reduces cross-domain discrepancy [8]. Text-driven semantic alignment and vision–language approaches use textual semantic guidance and cross-modal alignment for cross-scene classification [14], [15], [16]. Frequency-aware methods employ frequency-domain enhancement, Fourier phase preservation, or Fourier-based spectral–spatial generation [17], [18], [19]. Other methods decouple and align modality-shared semantics, progressively enhance domain-invariant features, perform two-stage domain alignment, or use dual-branch alignment [20], [21], [22], [23].

Despite these advancements, some DG methods adopt relatively complex domain-expansion or generative architectures [24]. Such designs can increase memory and computational demands, limiting their practicality on platforms with restricted resources, for example, UAVs and onboard processors. Consequently, there is a need for DG frameworks that maintain spectral–spatial discriminability with lower parameter complexity.

To address these challenges, we propose the adaptive basis spectral network (ABSNet), a lightweight single-source DG (SDG) framework tailored for HSI classification. ABSNet dynamically constructs sample-specific low-rank adaptation matrices from a compact set of learnable spectral bases, supporting adaptation to nonlinear spectral variations with minimal parameter overhead. Coupled with a spectral–spatial collaborative tokenizer (SSCT) and efficient token fusion strategies, the proposed framework learns discriminative representations for cross-domain classification while using only a few thousand trainable parameters. Experiments on the

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Houston, Pavia, and HyRANK datasets show that ABSNet achieves performance comparable to state-of-the-art (SOTA) methods with substantially fewer parameters, indicating its potential for resource-constrained deployment.

To sum up, the main contributions of this article are summarized as follows.

1) We propose the ABSNet, which adopts a dynamic adapter generation mechanism to construct sample-specific, rank-constrained adaptation matrices from a compact set of low-rank spectral bases. This mechanism supports adaptation to nonlinear domain shifts with minimal parameter overhead.
2) We propose a dual-branch tokenization framework that combines collaborative gating with low-rank adaptation. This mechanism provides complementary spatial and instance-adaptive token streams for compact feature embedding with minimal parametric overhead.
3) We conduct experiments on the Houston, Pavia, and HyRANK datasets. The results show that ABSNet delivers performance comparable to SOTA methods with substantially fewer parameters, indicating competitive generalization performance and potential suitability for resource-constrained deployment.

The rest of this article is organized as follows. Section II reviews related work on DG and lightweight methods for HSI classification. Section III details the proposed ABSNet. Section IV presents the experimental settings, ablation studies, and comparisons on the Houston, Pavia, and HyRank datasets. Section V concludes this article and discusses future research directions.

II. Related Work

A. DG for HSI Classification

Due to the extremely high cost of pixel-level annotation in remote sensing, labeled hyperspectral data are typically available from only one or a few source scenes. Conventional supervised models trained on such limited source domains often show noticeable performance degradation when applied to unseen target areas with different sensors, acquisition conditions, or land-cover distributions. DG has therefore become an important research direction, seeking to learn domain-invariant representations using solely source-domain labels. DG approaches can generally be divided into SDG, which relies on only one labeled domain and represents a particularly challenging yet highly realistic scenario, and multisource DG (MDG), which benefits from multiple heterogeneous source domains but remains less common in hyperspectral practice due to annotation scarcity.

Recent SDG methods for hyperspectral classification have explored various strategies to achieve robust invariant spectral–spatial representations. For instance, SDENet [11] incorporates supervised contrastive learning within a semantic encoder to promote intraclass compactness across domains. LDGNet [16] leverages large-scale vision–language priors through a dual-stream architecture to improve visual-semantic consistency, while involving a relatively large number of parameters. TSDAnet [22] explores a two-stage strategy for cross-scene hyperspectral classification. These approaches have contributed meaningfully to enhancing cross-scene robustness. At the same time, many current methods still rely on comparatively complex backbones or fixed projection designs, which may limit their flexibility in addressing strongly nonlinear and instance-level spectral variations. There thus remains valuable potential to develop more parameter-efficient solutions capable of delivering competitive generalization performance under significant domain shifts while better supporting practical deployment on resource-constrained platforms.

B. Lightweight HSI Classification

Deploying hyperspectral classification models on edge devices, such as uncrewed aerial vehicles and satellite onboard processors, imposes strict requirements on model size and computational complexity to enable real-time processing within limited power and memory budgets. Conventional deep learning architectures, particularly those based on Transformers for DG, typically involve a large number of parameters and substantial inference costs, which can limit their practicality in such resource-constrained environments.

Recent developments in lightweight HSI classification have introduced several promising parameter-efficient design strategies. For example, MADANet [25] integrates multiscale feature aggregation with dual channel-spatial attention modules to achieve a favorable balance between representation capacity and efficiency. L3DDAN [26] employs a compact 3-D dense autoencoder that reuses intermediate features through dense connections, effectively reducing redundancy during spectral–spatial processing. LiteDepthwiseNet [27] adopts depthwise separable convolutions combined with lightweight residual blocks to extract discriminative features with low computational overhead. CSM-DSPCss-Ghost [28] further explores dynamic split pointwise convolution and Ghost-style lightweight blocks, enabling parameter and computational reduction while retaining discriminative spectral–spatial features. In the context of DG, ACB [29] is an early lightweight SDG method, utilizing an attention-guided cascaded bottleneck structure that progressively refines features through compact multihead attention units.

While these parameter-efficient approaches have demonstrated encouraging results, many of them apply relatively uniform feature transformations or rely on fixed spectral priors across all samples. On the contrary, the proposed ABSNet adopts a sample-adaptive, low-rank adaptation mechanism that dynamically tailors spectral compensation for each input hyperspectral sample. This design provides sample-specific adaptation with a considerably smaller number of adaptation parameters, without depending on hand-crafted priors.

C. Low-Rank Representation in HSI Analysis

The high dimensionality of hyperspectral data is inherently accompanied by strong interband correlation and redundancy, meaning that observed spectra often approximately reside in a low-dimensional subspace spanned by a limited number of pure materials or endmembers. This fundamental low-rank

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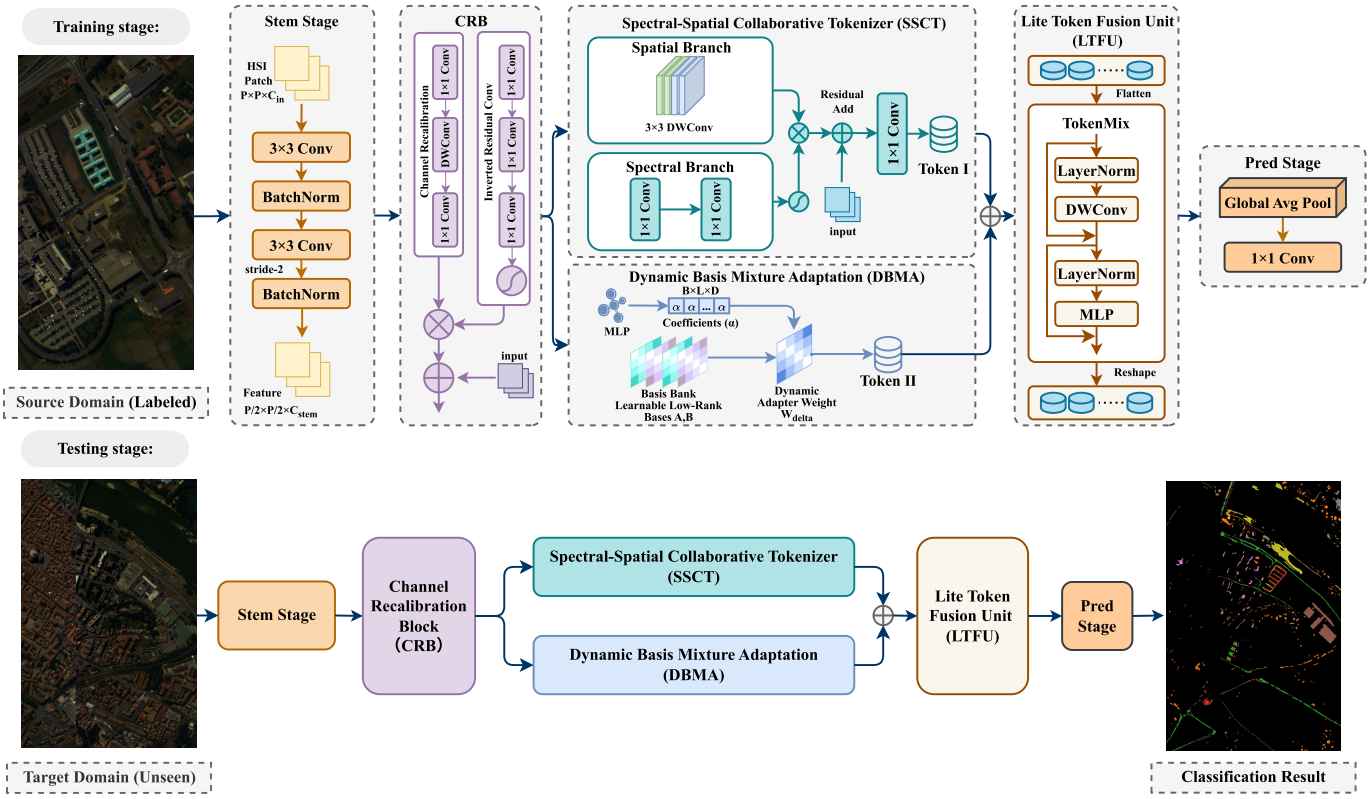


Fig. 1. Overall architecture of ABSNet. Dynamic low-rank adaptation is combined with spectral–spatial collaborative tokenization and an efficient token fusion unit for lightweight domain-generalized HSI classification. Only the labeled source-domain pathway is used for training.

property has been widely utilized in classical hyperspectral processing tasks, such as spectral unmixing [30], [31], denoising [32], [33], restoration [34], and subspace clustering [35]. Recent deep learning studies have incorporated low-rank structure through tensor factorization [36] and adaptive low-rank tensor representations [37]. Related work has used spectral low-rank priors for self-supervised reconstruction [38] and adaptive or prompt-guided low-rank modeling for HSI analysis [39], [40].

While these techniques have been applied effectively to in-domain tasks, most assume a relatively static, scene-wide low-rank structure, limiting their flexibility in handling instance-level spectral variations from cross-domain shifts in illumination, atmosphere, or sensor characteristics. Recent work has also explored dynamic low-rank and sparse priors for HSI analysis [41], yet their application to domain-generalized hyperspectral classification remains underexplored. The proposed framework contributes to this direction by employing a compact set of learnable low-rank spectral bases combined with a lightweight per-sample modulation mechanism, aiming to adapt to nonlinear domain shifts while remaining compatible with the low-rank structure of hyperspectral signals and incurring minimal parameter overhead.

III. Proposed Method

A. Overall Architecture

The proposed ABSNet is designed to learn compact HSI representations for cross-domain classification with minimal parametric overhead. As illustrated in Fig. 1, the architecture follows a four-stage pipeline that combines local feature processing with adaptive token modulation.

The data flow begins with an input HSI patch $\mathbf{X} \in \mathbb{R}^{B\times C_{\text{in}}\times P\times P}$. The hierarchical stem stage first performs a spectral–spatial bottleneck transformation to map raw pixels into a latent space $\mathbf{X}_{stem} \in \mathbb{R}^{B\times C_{\text{stem}}\times P/2\times P/2}$. This stage reduces spatial redundancy while expanding the receptive field for subsequent processing. The subsequent channel recalibration block (CRB) performs a further stride-2 transformation, producing features $\mathbf{Y} \in \mathbb{R}^{B\times C_{\text{stage}}\times P/4\times P/4}$ for the tokenizer stage. By modeling interchannel dependencies, the CRB adaptively rescales feature maps before tokenization.

The refined features are then fed into the dual-branch Tokenizer Stage. Here, the data flow splits into two parallel streams: the SSCT produces an input-conditioned spectral–spatial token stream, while the dynamic basis mixture adaptation (DBMA) branch generates instance-specific adaptive tokens. These complementary streams are fused to form the token sequence $\mathbf{T}$. Finally, the Lite Token Fusion Unit (LTFU) refines the fused tokens through lightweight patch-level token-context fusion before global average pooling and classification.

B. Hierarchical Feature Extraction and Recalibration

Hyperspectral imagery involves a fundamental tradeoff between rich spectral information and structural redundancy. To learn compact representations for cross-domain classification, ABSNet employs a hierarchical feature extraction pipeline. The initial stage functions as a spectral–spatial

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bottleneck, progressively mapping the high-dimensional input $\mathbf{X}$ into a latent feature space $\mathbf{Y}$ through successive strided convolutions. This hierarchical downsampling expands the receptive field and alleviates the computational burden for subsequent tokenization.

To further refine this representation, we propose a CRB that combines an inverted residual transformation with channel recalibration. The recalibration follows the classic squeeze-and-excitation (SE) paradigm [42], while the expand–depthwise–project layout is inspired by MobileNetV2 [43]. This layout is adopted as a parameter-efficient alternative to a conventional residual block. In particular, the 1×1 expansion, depthwise spatial filtering, and 1×1 projection separate channel transformation from spatial filtering with a compact parameter budget. The expansion operation alone does not establish improved modeling of interband correlations or domain invariance; rather, it provides an efficient feature transformation before recalibration. Let $\mathbf{U}$ denote the features generated by this transformation. We derive a channelwise importance vector $\mathbf{s}$ by aggregating global spatial context

$$\mathbf{s} = \sigma\left(\mathcal{G}_{ex}\left(\delta\left(\mathcal{G}_{re}\left(\mathrm{GAP}\left(\mathbf{U}\right)\right)\right)\right)\right) \tag{1}$$

$$\mathbf{Y} = \left(\mathbf{U} \odot \mathbf{s}\right) + \mathcal{H}\left(\mathbf{X}_{in}\right) \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid activation, $\delta(\cdot)$ is the ReLU activation function, GAP$(\cdot)$ denotes the global average pooling, $\mathcal{G}_{re}(\cdot)$ and $\mathcal{G}_{ex}(\cdot)$ represent the dimensionality reduction and expansion projections, respectively, $\odot$ denotes the elementwise multiplication, and $\mathcal{H}(\cdot)$ is the residual mapping from the input feature $\mathbf{X}_{in}$. By modulating $\mathbf{U}$ with $\mathbf{s}$, the CRB adaptively rescales the output feature maps before tokenization.

C. Dual-Branch Robust Tokenization

A critical challenge in lightweight HSI classification is handling instance-level spectral shifts caused by varying illumination or sensor characteristics. Fixed linear projections often lack the capacity to accommodate such nonstationary distributions. To address this, we propose a dual-branch tokenizer that combines an input-conditioned spectral–spatial component with a dynamic adaptive component.

1) Spectral–Spatial Collaborative Tokenizer: The SSCT branch generates an input-conditioned spectral–spatial token representation. Given an input feature map $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$, a 3×3 depthwise convolution performs local spatial filtering independently for each channel

$$\mathbf{F}_{\mathrm{spat}} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{DWConv}_{3\times 3}\left(\mathbf{F}\right)\right)\right), \mathrm{groups} = C. \tag{3}$$

In parallel, a lightweight two-layer 1×1 convolutional branch predicts an input-conditioned, spatially varying channel-gating map $\mathbf{A} \in \mathbb{R}^{H\times W\times C}$

$$\mathbf{A} = \sigma\left(\mathrm{Conv}_2\left(\mathrm{ReLU}\left(\mathrm{Conv}_1\left(\mathbf{F}\right)\right)\right)\right) \tag{4}$$

where Conv_1 reduces the channel dimension by a factor of 4 and Conv_2 restores it to C. Because the 1×1 convolutions operate at each spatial location, $\mathbf{A}$ is not a static channel-weighting vector. The token generation process is formulated as

$$\mathbf{T}_{ssct} = \mathcal{P}\left(\left(\mathbf{F}_{\mathrm{spat}} \odot \mathbf{A}\right) + \mathbf{F}\right) \tag{5}$$

where $\mathcal{P}(\cdot)$ represents a linear projection layer $\mathbf{W}_p \in \mathbb{R}^{C\times d}$ that maps the fused features into the token dimension d. The residual connection retains the ungated input features, while the gating branch provides input-dependent reweighting of the local spatial responses.

2) Dynamic Basis Mixture Adaptation: To capture sample-specific spectral variations without incurring the heavy parameter cost of standard dynamic networks, the DBMA branch introduces a sample-specific, low-rank adaptation mechanism. Accordingly, we construct a set of M shared low-rank basis pairs $\{\mathbf{A}_m \in \mathbb{R}^{d\times r}, \mathbf{B}_m \in \mathbb{R}^{r\times C}\}$, where each pair defines a rank-r ($r \ll C, d$) transformation expert. For each input instance, the network infers a mixing coefficient vector $\mathbf{c} \in \mathbb{R}^M$ based on the global statistics

$$\mathbf{W}_{\Delta}^{(b)} = \sum_{m=1}^{M} c_m^{(b)}\left(\mathbf{A}_m\mathbf{B}_m\right), \quad \mathbf{c} = \mathrm{Softmax}\left(\mathrm{MLP}\left(\mathrm{GAP}\left(\mathbf{Y}\right)\right)\right) \tag{6}$$

$$\mathbf{T}_{\Delta} = \mathbf{Y}_{flat}\left(\mathbf{W}_{\Delta}^{(b)}\right)^{T}. \tag{7}$$

Here, $\mathbf{Y}_{flat} \in \mathbb{R}^{N\times C}$ is the spatially flattened feature map. The term $\mathbf{A}_m\mathbf{B}_m$ represents the matrix product forming a basis matrix. By linearly combining these bases, the model can capture aspects of nonlinear spectral variation. The final token representation $\mathbf{T} \in \mathbb{R}^{N\times d}$ is a fusion of the two streams

$$\mathbf{T} = \mathbf{T}_{ssct} + \alpha\mathbf{T}_{\Delta} \tag{8}$$

where $\alpha = 0.4$ is a fixed residual scaling factor for adaptive residual adjustment.

D. Sequential Token Fusion and Prediction

The LTFU refines the fused patch-level token grid through a 5×5 depthwise TokenMix operation and an MLP

$$\mathbf{T}_{mid} = \mathbf{T} + \beta \cdot \mathcal{F}_{spat}\left(\mathrm{LN}\left(\mathbf{T}\right)\right) \tag{9}$$

$$\mathbf{T}' = \mathbf{T}_{mid} + \beta \cdot \mathcal{F}_{chan}\left(\mathrm{LN}\left(\mathbf{T}_{mid}\right)\right) \tag{10}$$

where LN$(\cdot)$ denotes layer normalization, $\mathcal{F}_{spat}$ and $\mathcal{F}_{chan}$ represent spatial and channel mixing functions implemented via depthwise convolutions and MLPs, respectively. For the default $P = 12$ input patch, the two stride-2 stages yield a 3×3 token grid; hence, the 5×5 TokenMix kernel covers the complete patch-level grid. The factor $\beta = 0.5$ scales the residual updates. Finally, the class logits $\mathbf{z} \in \mathbb{R}^K$ for K classes are generated via

$$\mathbf{z} = \mathrm{Flatten}\left(\mathbf{W}_{cls} * \mathrm{GAP}\left(\mathbf{T}'\right)\right) \tag{11}$$

where $\mathbf{W}_{cls}$ denotes the weights of the final 1×1 convolutional classifier and $*$ represents the convolution operation.

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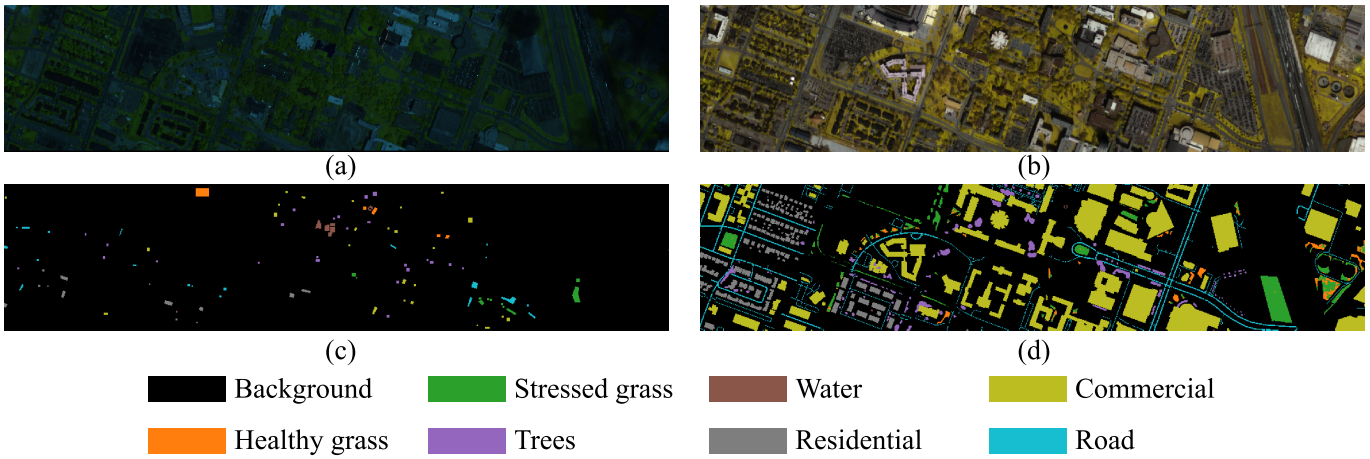


Fig. 2. Pseudocolor images and ground-truth maps of the Houston dataset. (a) Houston 2013 pseudocolor. (b) Houston 2018 pseudocolor. (c) Houston 2013 ground truth. (d) Houston 2018 ground truth.

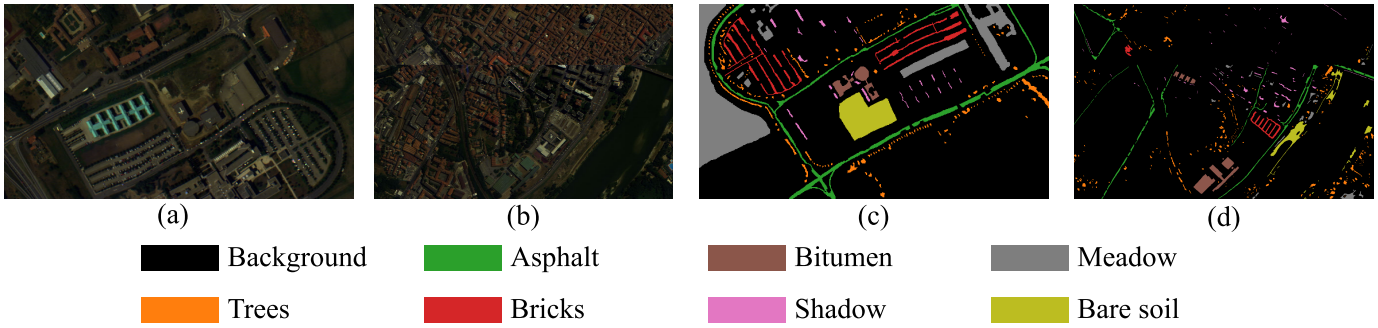


Fig. 3. Pseudocolor images and ground-trutth maps of the Pavia dataset. (a) Pavia University pseudocolor. (b) Pavia Center pseudocolor. (c) Pavia University ground truth. (d) Pavia Center ground truth.

IV. Experiments

A. Datasets

1) Houston Dataset: The Houston dataset consists of two HSIs acquired over the University of Houston campus in 2013 and 2018, respectively. Despite covering the same geographical area, the two acquisitions exhibit significant spectral shifts due to different sensors and imaging conditions. Following the standard SDG protocol [11], [29], the 2013 image is cropped and reduced to 48 bands to match the 2018 image, retaining seven shared land-cover classes. Houston 2013 serves as the labeled source domain, and Houston 2018 is reserved as the unseen target domain for final evaluation. Class statistics are listed in Table I, with pseudocolor images and ground-truth maps shown in Fig. 2.

TABLE I
Class Distribution in the Houston Dataset

No.	Class name	Houston13	Houston18
1	Healthy grass	345	1353
2	Stressed grass	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

2) Pavia Dataset: The Pavia dataset consists of two hyperspectral scenes acquired over the city of Pavia, Italy, in 2001. The first scene is named Pavia University and measures 610×340 pixels with 103 spectral bands. The second scene is named Pavia Center and measures 1096×715 pixels with 102 spectral bands. For spectral consistency, the University scene is reduced to 102 bands and seven shared land-cover classes are retained. Following the standard SDG protocol, Pavia University serves as the labeled source domain, and Pavia Center is reserved as the unseen target domain for final evaluation. Class statistics are listed in Table II. Pseudocolor images and ground-truth maps are shown in Fig. 3.

3) HyRank Dataset: The HyRank dataset consists of two hyperspectral scenes acquired over rural areas in Greece. Dioni measures 250×1376 pixels with 176 spectral bands, while Loukia measures 246×954 pixels with the same number of bands. Although captured at different times and locations, the two scenes share 12 land-cover classes. Following established SDG benchmarks [29], [44], Dioni serves as the labeled source domain, and Loukia is reserved as the unseen target domain for final evaluation. Class statistics are listed in Table III,

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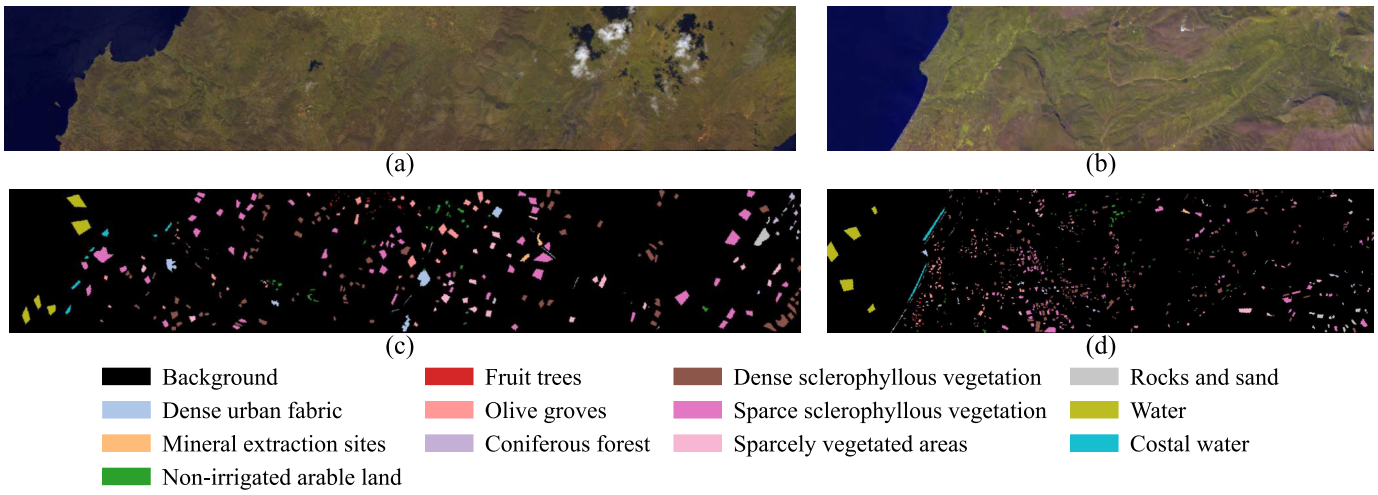


Fig. 4. Pseudocolor images and ground-truth maps of the HyRank dataset. (a) Dioni pseudocolor. (b) Loukia pseudocolor. (c) Dioni ground truth. (d) Loukia ground truth.

TABLE II
CLASS DISTRIBUTION IN THE PAVIA DATASET

No.	Class name	Pavia University	Pavia Center
1	Trees	3064	7598
2	Asphalt	6631	9248
3	Bricks	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

TABLE III
CLASS DISTRIBUTION IN THE HYRANK DATASET

No.	Class name	Dioni	Loukia
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non-irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sclerophyllous vegetation	5035	2996
8	Sparse sclerophyllous vegetation	6374	2361
9	Sparcely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
Total		20024	10317

and the pseudocolor images and ground-truth maps are shown in Fig. 4.

B. Implementation Details

All experiments are conducted using PyTorch with an NVIDIA GeForce RTX 4070 Ti GPU and a batch size of 256. The experiments are repeated using different random seeds to assess result stability. Training relies exclusively on labeled source-domain samples. Data augmentation consists of random horizontal/vertical flipping and radiation augmentation with an enhancement amplitude ratio of 3. The model is trained for 300 epochs using the Adam optimizer with an initial learning rate of 10^{-2}. Supervision is provided solely on the source domain through the negative log-likelihood loss

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\log \hat{y}_{i,y_i} \tag{12}$$

where $\hat{y}_i$ denotes the predicted probability of the ground-truth class for the ith sample. During training, hyperparameter tuning, and model selection, only labeled source-domain data are used. The target domain is reserved exclusively for final evaluation.

The classification objective provides end-to-end discriminative supervision to all learnable components of ABSNet. We retain this compact source-supervised formulation to isolate the effect of the parameter-efficient feature transformations. This design is consistent with the lightweight SDG formulation adopted by ACB [29].

Classification performance is assessed using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} n_{ii}}{N_{\text{total}}} \times 100\% \tag{13}$$

$$\mathrm{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{n_{ii}}{n_{i\cdot}} \times 100\% \tag{14}$$

$$\mathrm{Kappa} = \frac{P_o - P_e}{1 - P_e} \times 100\% \tag{15}$$

where n_{ii} represents the correctly classified pixels of class i, N_{total} is the total number of target-domain pixels, $P_o = \sum_{i=1}^{C} n_{ii}/N_{\text{total}}$ denotes the observed agreement, and $P_e = \sum_{i=1}^{C} n_{i\cdot}n_{\cdot i}/N_{\text{total}}^2$ denotes the chance agreement.

C. Experiment Results

To assess the effectiveness and efficiency of ABSNet, we compare it with recent DG approaches, including SDENet

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON HOUSTON DATASET (%). BEST RESULTS ARE IN BOLD

Class	GhostNet [46]	SDENet [11]	LLURNet [24]	LDGNet [16]	ACB [29]	D3Net [13]	EHSNet [45]	ISDGS [44]	TSTNet [47]	Ours
1	15.14	44.89	48.27	48.75	44.46	39.19	62.36	78.20	62.49	30.05
2	59.67	77.85	82.73	73.11	73.40	75.25	74.09	75.62	71.18	70.63
3	53.87	59.99	63.26	64.41	66.47	69.05	69.58	62.76	60.62	54.49
4	27.53	100.00	100.00	81.55	52.88	27.79	42.21	100.00	11.01	25.04
5	62.70	60.06	69.18	64.26	64.09	63.78	61.97	70.81	56.42	59.94
6	87.25	90.02	73.80	90.02	89.61	87.61	89.48	89.27	85.19	89.42
7	52.34	54.21	61.97	52.72	59.26	60.16	63.34	48.00	54.03	58.36
OA (%)	75.41 ±0.51	79.03 ±0.95	77.29 ±1.40	80.14 ±0.88	**80.40 ±0.10**	79.26 ±0.16	80.45 ±0.19	79.52 ±1.17	74.38 ±0.16	77.91 ±0.72
AA (%)	57.16 ±4.79	70.80 ±2.34	73.88 ±0.93	67.76 ±2.89	66.82 ±3.28	68.11 ±2.61	70.24 ±1.48	**74.95 ±3.24**	74.11 ±1.52	62.74 ±3.08
Kappa (%)	57.15 ±2.18	64.39 ±2.38	62.28 ±2.13	65.30 ±2.01	65.17 ±0.17	63.81 ±0.89	**66.58 ±0.54**	65.30 ±3.21	59.44 ±1.30	62.74 ±0.15
Params	5.03K	1.83M	564.55k	34.22M	17.32k	2.19M	33.86M	634.17k	5.33M	**3.74k**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON PAVIA DATASET (%). BEST RESULTS ARE IN BOLD

Class	GhostNet [46]	SDENet [11]	LLURNet [24]	LDGNet [16]	ACB [29]	D3Net [13]	EHSNet [45]	ISDGS [44]	TSTNet [47]	Ours
1	84.06	85.33	84.76	86.27	89.52	81.03	87.46	87.10	69.74	91.42
2	80.96	83.49	86.72	84.38	83.17	84.00	89.35	87.90	88.28	86.42
3	58.95	61.15	70.61	53.28	62.46	69.14	73.81	63.47	51.75	71.23
4	76.93	82.84	85.69	79.54	81.76	83.96	84.74	87.93	78.06	88.73
5	85.11	83.74	85.41	91.38	98.49	86.62	91.40	91.49	90.86	96.95
6	67.91	78.84	72.84	70.77	69.29	59.91	67.03	68.70	58.30	73.11
7	85.80	82.98	84.39	80.40	87.72	87.49	87.63	86.57	85.38	89.82
OA (%)	79.00 ±4.19	81.77 ±0.56	83.43 ±1.06	80.21 ±0.99	83.13 ±0.52	80.68 ±0.43	84.89 ±0.30	84.19 ±0.11	77.21 ±0.55	**87.00 ±0.34**
AA (%)	78.19 ±1.49	79.76 ±0.85	81.40 ±0.46	78.00 ±1.59	83.15 ±0.82	80.01 ±0.52	84.44 ±0.98	82.33 ±1.22	76.51 ±0.32	**85.97 ±0.56**
Kappa (%)	74.86 ±4.75	78.09 ±0.68	80.05 ±1.22	76.18 ±1.17	79.79 ±0.68	76.85 ±0.50	81.87 ±0.35	81.00 ±0.19	72.78 ±0.59	**84.34 ±0.40**
Params	8.83K	2.47M	608.47k	35.10M	19.39k	2.83M	34.47M	665.44k	5.34M	**5.71k**

[11], LLURNet [24], LDGNet [16], ACB [29], D3Net [13], EHSNet [45], and ISDGS [44]. In addition, an ultralightweight supervised baseline, GhostNet [46], is constructed by applying the official width multiplier of 0.15 to the original GhostNet to rule out capacity effects. TSTNet [47] is retained as a DA reference, following the evaluation practice of previous lightweight DG studies for cross-scene HSI classification [29]. The DG baselines and GhostNet are reimplemented under their respective source-only settings, whereas TSTNet follows its original DA protocol. Tables IV–VI show the classification performance of different methods on the three datasets. Figs. 5– 7 show the classification maps.

ABSNet achieves the highest OA, AA, and Kappa on Pavia and HyRank, while delivering competitive performance on Houston. It uses 3.74k–8.43k parameters, which is more than two orders of magnitude fewer than the Transformer-based competitors and approximately 65%–78% fewer than ACB. In Houston, ABSNet does not achieve the highest OA. The heterogeneous urban structures, shadows, and illumination conditions may require richer contextual modeling than the lightweight architecture can provide. Nevertheless, ABSNet has the lowest computational complexity and the fastest training time on this dataset among the compared methods, indicating a favorable balance between generalization performance and efficiency.

The classification maps provide qualitative visual comparisons. On Houston, ABSNet produces relatively sharp boundaries for large-area classes such as grass, water, and buildings. On Pavia, its predicted regions are generally more continuous for small objects. On HyRank, ABSNet preserves relatively complete object structures and class consistency, including those for several minority classes.

Table VII reports the average per-epoch training time and computational complexity of all compared methods. Although the training time of ABSNet is not always the fastest among lightweight competitors, its computational complexity is consistently among the lowest. When inference complexity, model size, and generalization accuracy are considered jointly, ABSNet provides a favorable accuracy–efficiency tradeoff among the compared methods. These results indicate its potential for deployment on resource-constrained platforms such as UAVs and satellite onboard processors.

D. Parameter Analysis

1) Impact of DBMA Hyperparameters: The expressive power and generalization behavior of the DBMA module

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TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON HYRANK DATASET (%). BEST RESULTS ARE IN BOLD

Class	GhostNet [46]	SDENet [11]	LLURNet [24]	LDGNet [16]	ACB [29]	D3Net [13]	EHSNet [45]	ISDGS [44]	TSTNet [47]	Ours
1	33.00	26.94	23.90	35.35	27.71	20.18	17.14	33.41	22.99	29.23
2	28.86	22.80	25.31	5.48	36.00	1.67	0.87	41.94	15.39	35.05
3	57.25	48.94	36.61	56.92	46.60	65.12	58.42	66.36	17.28	53.95
4	21.27	9.84	3.27	16.98	14.89	21.29	9.18	5.70	0.00	5.63
5	0.96	1.16	3.84	0.18	31.45	22.07	6.87	0.35	66.87	74.35
6	42.50	33.67	32.27	45.00	24.89	41.78	32.88	36.35	16.74	1.50
7	66.17	75.20	73.56	66.46	68.02	70.77	66.62	69.37	72.15	73.00
8	59.00	68.02	71.72	59.19	61.00	61.21	60.35	67.33	58.07	59.69
9	36.93	47.39	31.56	35.03	17.91	32.27	33.80	29.18	34.63	38.27
10	0.58	6.08	7.12	0.00	7.03	13.63	19.05	2.28	44.91	28.03
11	99.99	100.00	99.94	100.00	99.96	99.99	99.78	99.36	100.00	99.99
12	93.50	100.00	97.35	99.88	90.35	88.26	94.78	91.22	96.34	92.96
OA (%)	60.84 ±0.28	61.34 ±0.94	62.45 ±1.36	60.83 ±0.63	62.52 ±0.65	63.10 ±1.04	61.36 ±0.90	64.23 ±0.79	65.07 ±1.08	**67.82 ±0.98**
AA (%)	43.86 ±2.30	45.73 ±1.59	43.31 ±1.71	46.33 ±1.18	44.15 ±2.03	46.84 ±1.99	42.42 ±0.95	46.07 ±1.24	46.89 ±0.47	**48.77 ±0.24**
Kappa (%)	51.34 ±0.39	51.93 ±0.84	52.94 ±1.39	51.52 ±0.79	52.16 ±0.70	54.67 ±1.17	51.77 ±0.84	55.23 ±0.67	57.49 ±1.17	**60.67 ±0.82**
Params	14.15K	3.33M	671.18k	36.27M	23.79k	3.7M	36.01M	710.8k	5.37M	**8.43k**

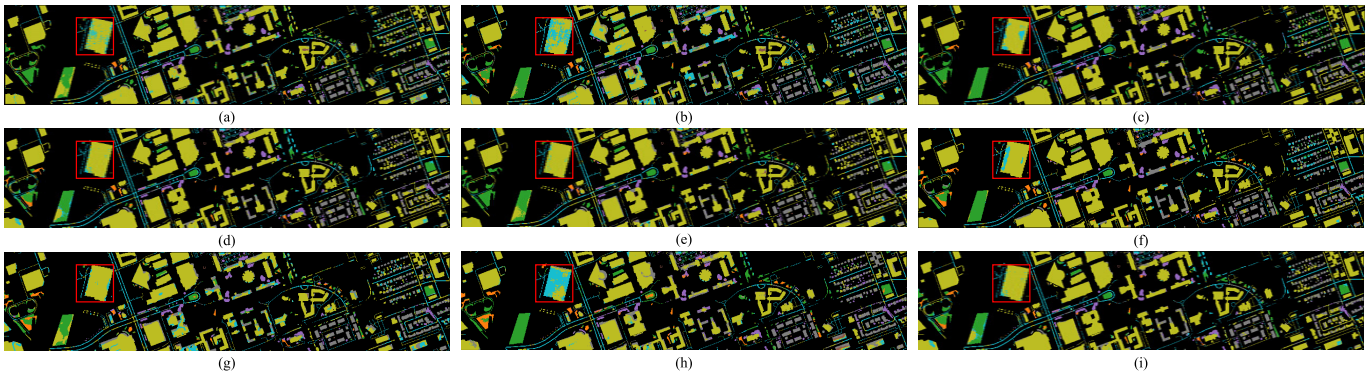


Fig. 5. Classification maps of different methods on the Houston dataset. (a) SDENet. (b) LLURNet. (c) LDGNet. (d) ACB. (e) D3Net. (f) EHSNet. (g) ISDGS. (h) TSTNet. (i) ABSNet.

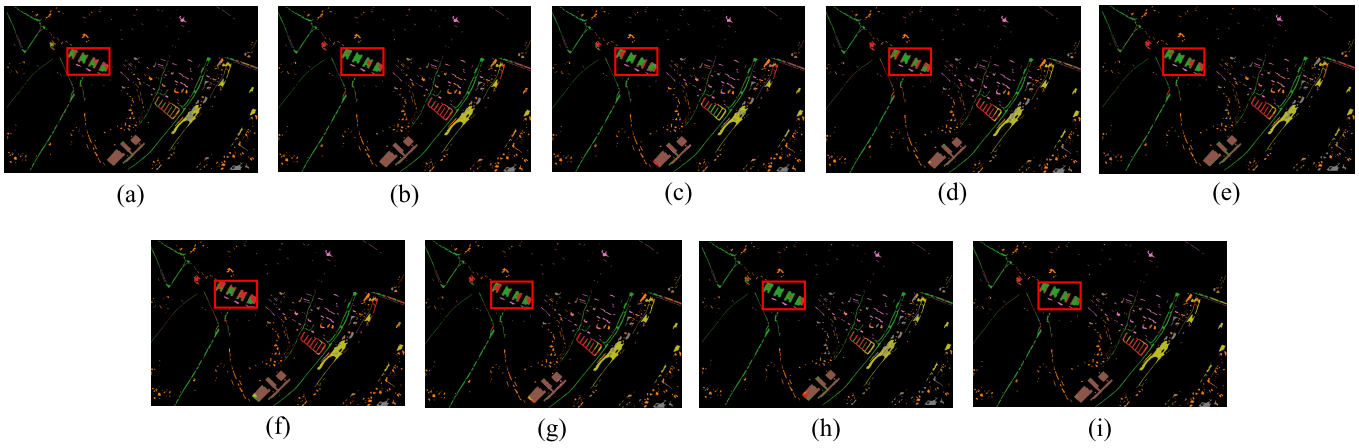


Fig. 6. Classification maps of different methods on the Pavia dataset. (a) SDENet. (b) LLURNet. (c) LDGNet. (d) ACB. (e) D3Net. (f) EHSNet. (g) ISDGS. (h) TSTNet. (i) ABSNet.

are primarily governed by two hyperparameters: the number of shared bases $M \in \{1, 2, 4, 8\}$ and the intrinsic rank $r \in \{1, 2, 4, 8\}$. We further report sensitivity analyses of these architectural hyperparameters. As illustrated in Fig. 8, moderate configurations ($M = 2$, $r = 1$) provide a favorable balance between adaptation flexibility and parameter efficiency in these analyses. Increasing either M or r introduces additional parametric capacity and is associated with lower performance

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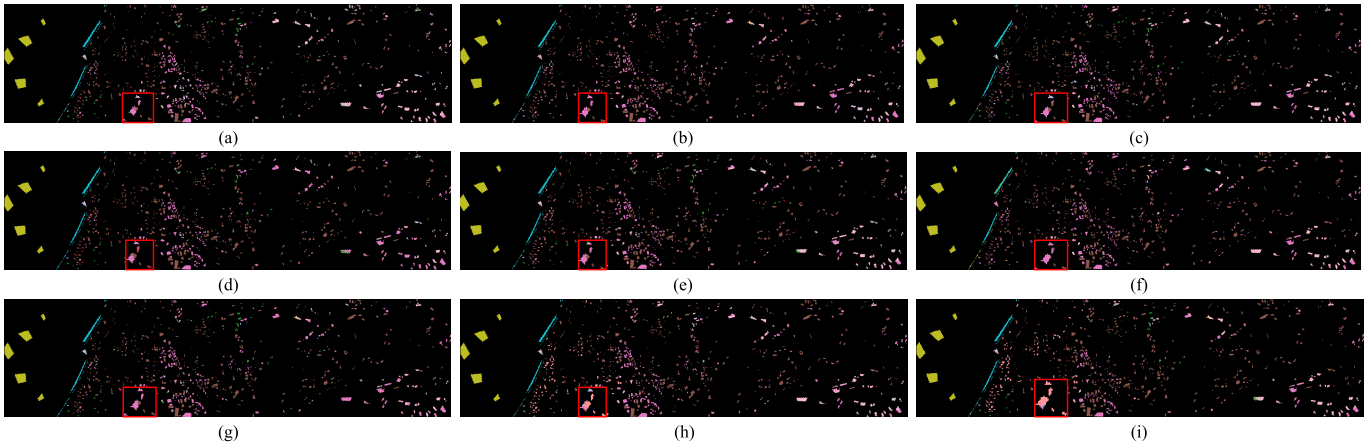


Fig. 7. Classification maps of different methods on the HyRank dataset. (a) SDENet. (b) LLURNet. (c) LDGNet. (d) ACB. (e) D3Net. (f) EHSNet. (g) ISDGS. (h) TSTNet. (i) ABSNet.

TABLE VII
TRAINING TIME PER EPOCH AND COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON THREE DATASETS

Method	Houston		Pavia		HyRank	
	Time	FLOPs	Time	FLOPs	Time	FLOPs
SDENet [11]	6.01s	4.34M	25.10s	8.10M	18.91s	13.26M
LLURNet [24]	5.43s	4.34M	24.47s	8.10M	18.24s	13.26M
LDGNet [16]	17.41s	41.86M	61.27s	89.42M	37.20s	153.51M
ACB [29]	0.76s	0.34M	8.87s	**0.47M**	5.29s	**0.64M**
D3Net [13]	12.24s	13.35M	45.84s	17.70M	24.41s	23.65M
EHSNet [45]	33.23s	64.60M	61.67s	206.12M	51.87s	518.95M
ISDGS [44]	3.91s	4.34M	12.67s	8.10M	6.60s	13.26M
TSTNet [47]	2.20s	8.50M	12.49s	10.06M	5.97s	12.19M
Ours	**0.72s**	**0.27M**	**8.32s**	0.55M	**4.70s**	0.94M

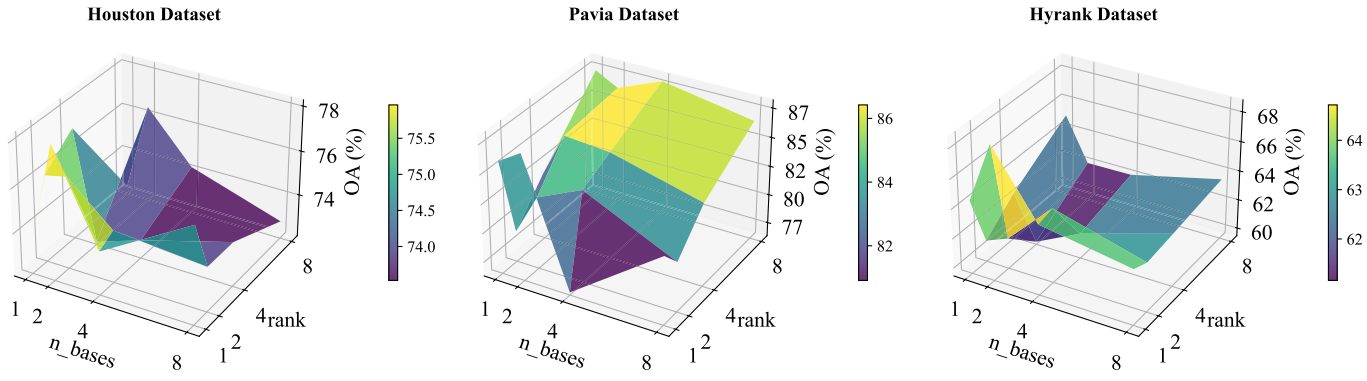


Fig. 8. Sensitivity to the number of bases and rank.

on the evaluated datasets, particularly Houston and HyRank. The configuration $M = 2$, $r = 1$ ranks among the better-performing settings while using the fewest parameters. These observations are consistent with the role of the low-rank constraint in limiting adaptation capacity.

2) Impact of Token Fusion Weight: The scalar weight α governs the relative contribution of DBMA-generated compensation tokens and the base tokens produced by the SSCT branch. We report the sensitivity to $\alpha \in \{0.1, 0.2, \ldots, 0.9\}$ in Fig. 9. The results show that $\alpha = 0.4$ lies in a stable, high-performing range, balancing the SSCT representation with moderate sample-specific corrections from DBMA.

3) Impact of LTFU Residual Scaling Factor: The coefficient β is the residual scaling factor in the LTFU. It controls the magnitude of the token-mixing and MLP updates relative to the input tokens in the two LTFU update equations. To examine its effect, we evaluate $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$

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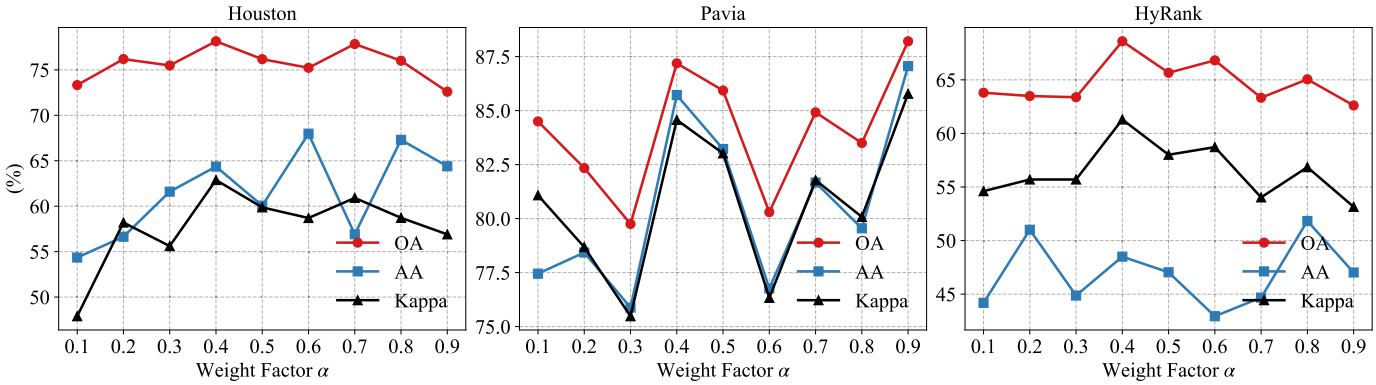


Fig. 9. Sensitivity to token fusion weight α.

TABLE VIII
ABLATION STUDY OF THE CONTRIBUTION OF EACH PROPOSED COMPONENT. BEST RESULTS ARE IN BOLD

Backbone	CRB	SSCT	DBMA	Houston (%)			Pavia (%)			HyRank (%)		
				OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
√				74.99	57.16	56.66	84.55	82.96	81.42	62.36	39.83	52.14
√	√			76.61	55.30	57.98	84.71	82.36	81.59	64.12	41.23	55.01
√	√	√		71.87	56.37	54.90	87.44	85.24	84.83	63.36	52.78	54.84
√	√		√	76.39	52.12	60.19	86.72	84.97	84.00	67.04	45.10	59.73
√	√	√	√	**78.15**	**64.36**	**62.88**	**87.19**	**85.72**	**84.57**	**68.60**	**48.50**	**61.30**

TABLE IX
ABLATION STUDY OF LTFU PLACEMENT. BEST RESULTS ARE IN BOLD

LTFU placement	Houston (%)			Pavia (%)			HyRank (%)		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
None	78.02	**73.06**	**63.65**	80.73	81.75	77.03	64.90	42.67	56.55
Early (before DBMA fusion)	75.81	58.38	59.32	86.21	**86.52**	83.44	63.84	43.00	54.93
Late (proposed)	**78.15**	64.36	62.88	**87.19**	85.72	**84.57**	**68.60**	**48.50**	**61.30**

while keeping all other settings unchanged. As summarized in Fig. 10, performance improves as β increases to 0.5, whereas larger values lead to gradual degradation. Small values limit the contribution of the LTFU refinement, while excessively large values overemphasize residual updates. Performance remains relatively stable for $\beta \in [0.3, 0.7]$. In this analysis, $\beta = 0.5$ provides a stable balance between input-token preservation and LTFU refinement.

E. Ablation Study

1) Component Contributions: To quantitatively assess the contribution of each component, ablation studies are conducted on the Houston, Pavia, and HyRank datasets under identical training conditions (see Table VIII). Removing the CRB degrades the overall performance, supporting its contribution to the full architecture. Adding SSCT to the CRB+DBMA configuration further strengthens the combined design, supporting a complementary role for SSCT and DBMA in the full model.

2) Impact of LTFU Placement: To examine whether the LTFU should operate before or after DBMA fusion, we compare three variants: no LTFU, early LTFU applied to the SSCT tokens before DBMA fusion, and the proposed late LTFU applied to the fused tokens. As shown in Table IX, the late placement provides the strongest overall results. This pattern supports contextualizing the fused local and adaptive tokens after token fusion, rather than applying LTFU before adaptation, under the present experimental setting.

3) Impact of CRB Structure: To assess the structural choice in the CRB, we compare the proposed inverted residual design with a matched conventional residual block. As reported in Table X, the inverted design yields higher classification accuracy across the three benchmark datasets while using fewer parameters. These results support its selection as a parameter-efficient option in ABSNet under the present experimental setting.

4) Impact of DBMA Design: To evaluate the effectiveness of the two core components in the DBMA module—low-rank decomposition and instancewise coefficient prediction—we designed three alternative adaptation strategies for comparison. Variant A removes the low-rank constraint and learns a complete transformation matrix. Variant B retains the low-rank decomposition but applies the same update to all samples without instance-specific modulation. Variant C maintains multiple low-rank bases while replacing the learned dynamic coefficients with fixed uniform weights. Quantitative results

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TABLE X
ABLATION STUDY OF DIFFERENT STRUCTURAL DESIGNS FOR THE CRB MODULE. BEST RESULTS ARE IN BOLD

Design	Houston (%)				Pavia (%)				HyRank (%)			
	OA	AA	Kappa	Params	OA	AA	Kappa	Params	OA	AA	Kappa	Params
Conventional residual	77.57	64.03	61.67	5.23K	84.37	78.25	81.00	7.13K	63.54	46.21	57.70	9.79K
Proposed CRB (Inverted)	**78.15**	**64.36**	**62.88**	**3.74K**	**87.19**	**85.72**	**84.57**	**5.71K**	**68.60**	**48.50**	**61.30**	**8.43K**

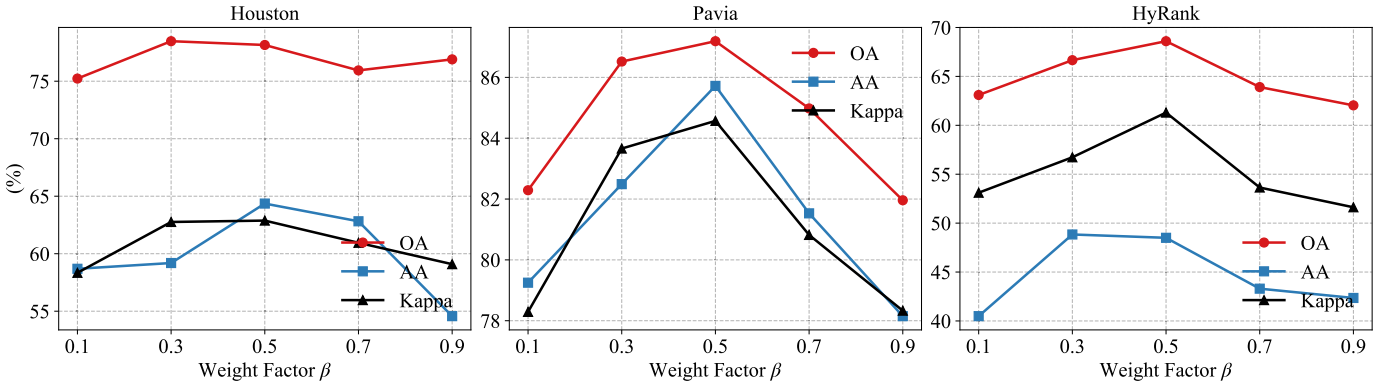


Fig. 10. Sensitivity to LTFU residual scaling factor β.

TABLE XI
ABLATION STUDY OF DIFFERENT SPECTRAL ADAPTATION STRATEGIES FOR THE DBMA MODULE. BEST RESULTS ARE IN BOLD

Strategy	Houston (%)			Pavia (%)			HyRank (%)		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Full-rank adapter	76.85	**70.07**	62.56	82.58	79.35	78.94	61.90	42.47	51.86
Standard LoRA	77.42	66.47	61.40	86.70	84.25	83.95	65.86	45.01	57.40
LoRA with uniform bases	75.45	57.43	57.61	80.81	80.09	77.04	64.55	47.65	56.78
Proposed DBMA	**78.15**	64.36	**62.88**	**87.19**	**85.72**	**84.57**	**68.60**	**48.50**	**61.30**

are summarized in Table XI. Across all three benchmark datasets, the proposed DBMA achieves the highest OA and Kappa. On the HyRank dataset, it improves over standard LoRA (Variant B) by 2.74 percentage points in OA and 3.49 percentage points in AA. Improvements are also observed on Houston and Pavia across the reported metrics. These results indicate that combining low-rank spectral bases with sample-adaptive coefficient prediction contributes to the DG performance of DBMA in the evaluated settings.

V. CONCLUSION

This article proposes ABSNet, a lightweight DG framework for HSI classification under cross-domain spectral shifts and resource constraints. By dynamically synthesizing sample-specific low-rank adaptation matrices from a compact set of shared spectral bases, ABSNet is designed to reduce the impact of nonlinear distribution discrepancies while retaining discriminative spectral information with minimal parameter overhead. The SSCT and LTFU support the learning of discriminative, compact representations. Experiments on the Houston, Pavia, and HyRank cross-scene benchmarks show that ABSNet achieves competitive generalization performance while using only a few thousand trainable parameters and maintaining a low training time and computational complexity. This accuracy–efficiency tradeoff suggests that ABSNet may be useful for resource-constrained platforms such as uncrewed aerial vehicles and satellite onboard processors. Future work will evaluate the framework on larger hyperspectral datasets and on edge devices under practical operating conditions.

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